# Signed permutation groups at the ends of alternating group constructions

Human Compression project

Author edition; distinct review pending

This note supplies uniform proofs for the higher odd-degree families in Fernandes–Leemans, arXiv:1810.12450v2, Theorems 4.8–4.10. The literal permutation families belong to that source. We describe their actual end subgroups, including the relations between their separate orbits. The argument uses the standard end criterion for string C-groups, the acquired restriction-kernel theorem, and the supported alternating-group lemma from our reviewed companion note *Finite end groups for alternating group constructions*. The earlier maximal-rank family, explicitly attributed to [14] in source Theorem 4.7, is retained as an input, not counted as a new result of this note.

All signs below take values in $\mathbb{F}_2$. We regard symmetric and alternating groups on named sets as supported permutation groups, fixing every other point. A C-marking has the string commutations and the intersection property for every pair of subsets of its marks.

## 1 Two elementary transfer principles

The acquired restriction-kernel theorem says the following. For a C-marking $(h_1, \dots, h_d)$, suppose only $h_1$ moves a point $a$. Put $P_j = \langle h_1, \dots, h_j \rangle$ and $X_j = a^{P_j}$, with $P_0 = 1$. Adjoining the pure transposition $(c\ a)$ at a new point preserves C precisely when

$$P_j \cap \mathrm{Sym}(X_{j-1}) \le P_{j-1} \quad (1 \le j \le d). \tag{1}$$

The first test is trivial. The group after adjoining the transposition is

$$\mathrm{Sym}(X_d \cup \{c\}) \times P_d^{\Omega \setminus X_d}. \tag{2}$$

The second factor is the actual restriction image. It follows by conjugating $(c\ a)$ to all star transpositions, then cancelling the action on $X_d$. Once the first mark is pure, another pure leaf may always be attached. We also use the proved forward endpoint twist: in a direct product with a central involution $z$, replacing an endpoint mark $h$ by $hz$ preserves C. No converse to this assertion is used.

**Lemma 1** (The quotient criterion)**.** *Suppose a homomorphism maps a string marking onto a C-marking, and is injective on one endpoint deletion of the original marking. Then the original marking is C.*

*Proof.* Reverse the ordering so that injection holds on deletion of the first mark. Induct on the length. That endpoint deletion is isomorphic, as a marked group, to its image and hence is C. The other endpoint deletion is C by induction, since its own deletion of the first mark is inside the subgroup on which the homomorphism is injective. An element in the intersection of the two endpoint groups maps into the middle group by the quotient's intersection property. Choose a word in the middle marks with that image. Both this word and the original element lie in the injective endpoint group, so they are equal. The end criterion proves the assertion. The length-one start is immediate. This is the usual McMullen–Schulte quotient criterion, Proposition 2E17. □

## 2 Two markings of the even signed permutation group

On $\{1^+, 1^-, \ldots, t^+, t^-\}$ let $s_i$ interchange the two coordinate pairs $i, i+1$, preserving their signs. Let $v_i$ interchange $i^+, i^-$. Thus

$$D_t = \{v_1^{e_1} \cdots v_t^{e_t} : \sum e_i = 0\} \rtimes S_t.$$

The symbol $D_t$ in this note denotes this concrete permutation group, of order $2^{t-1}t!$, rather than a dihedral group.

There are two strings. The first, for $t \geq 5$, is

$$s_1, \ldots, s_{t-4}, \quad T = s_{t-3}v_{t-1}v_t, \quad s_{t-2}, \quad s_{t-1}. \tag{3}$$

The second, for $t \geq 7$, is

$$s_1, \ldots, s_{t-6}, \quad T = s_{t-5}v_{t-3}v_{t-2}v_{t-1}v_t, \quad s_{t-4}, \quad s_{t-3}s_{t-1}, \quad s_{t-2}. \tag{4}$$

Empty initial lists have their usual meaning. In either case write $f$ for the displayed product of $v_i$ in $T$, and $p$ for its unsigned index: $p = t-3$ or $p = t-5$. Directly,

$$(s_{p-1}T)^3 = f. \tag{5}$$

Indeed the unsigned product is a three-cycle, and $f$ is disjoint from its three coordinates. Consequently the prefix ending in $T$ contains $f$ and every unsigned $s_i$ for $i \leq p$.

**Lemma 2.** *Both strings are C-markings of $D_t$. For fresh points $c, d$, one may prepend $(c\ 1^+)$, then prepend $(d\ c)$. After the two attachments the group is $S_{2t+2}$. In (3), deletion of the last mark after these attachments gives $S_{2t} \times C_2$ on a set of size $2t$ and the last signed pair. In (4), it gives $S_{2t-2} \times V_4$, where $V_4$ is the regular four-point group on the last two signed pairs.*

*Proof.* We first record the prefix groups before proving C. The prefix ending in $T$ is

$$N_t = \operatorname{diag} S_{p+1} \times \langle f \rangle. \tag{6}$$

Here the symmetric group acts on the first $p+1$ signed pairs without sign changes; $f$ has disjoint support. On adjoining $s_{p+1}$, the prefix becomes

$$M_t = \{v_1^{e_1} \cdots v_{p+2}^{e_{p+2}} (v_{p+3} \cdots v_t)^{\sum e_i} : e_i \in \mathbb{F}_2\} \rtimes S_{p+2}. \tag{7}$$

Conjugates of $f$ under the unsigned $S_{p+2}$ give each vector $v_i v_{p+3} \cdots v_t$, proving the equality. This is the usual signed symmetric group on the first $p+2$ coordinates, with every remaining sign recording the total number of their flips.

For (3), $M_t$ is the penultimate prefix. The final unsigned bridge gives $S_t$. The conjugates of $f$ give all even sign changes: subtracting two conjugates which differ at just two coordinates gives $v_i v_j$, and these vectors span the even subspace. Thus the full group is $D_t$.

For (4) there is one more intermediate prefix. Adjoin $b = s_{t-3}s_{t-1}$ to (7). Its group is

$$L_t = D_{t-2} \times \langle v_{t-1}v_t, s_{t-1} \rangle = D_{t-2} \times V_4. \tag{8}$$

To see that the unsigned factors separate, take the preceding unsigned transposition $a = s_{t-4}$. The cube $(ab)^3$ is $s_{t-1}$; hence $s_{t-3}$ is also available. We have unsigned $S_{t-2} \times S_{\{t-1,t\}}$. Its conjugates of $f$ are $v_i v_j v_{t-1} v_t$ with $i, j \leq t-2$. The product for the three pairs of a triangle gives $v_{t-1}v_t$ alone. Removing this factor gives all even flips on the first $t-2$ coordinates. These are exactly the independent factors in (8). The final bridge $s_{t-2}$ gives unsigned $S_t$, and the same even-vector argument gives the full $D_t$.

All distant marks commute by inspection. For C, the two finite starts are (3) at $t = 5$ and (4) at $t = 7$. The certificate checks all $16^2 = 256$ and $32^2 = 1024$ ordered pairs of subset subgroups, respectively, as actual permutation groups. These are finite premises; their extension is the following induction.

The prefix $N_t$ is an endpoint central twist of the ordinary adjacent-transposition simplex and is C. At every subsequent prefix, deleting its first mark gives the corresponding prefix with $t - 1$ coordinates, fixing the first signed pair pointwise. Deleting its last mark gives the preceding prefix in (6)–(8). The intersection is exactly the latter prefix's pointwise stabilizer of the first signed pair. Equations (6)–(8) show that this stabilizer is the preceding prefix with $t - 1$ coordinates: the unsigned permutation fixes coordinate one, and its sign there is zero; the same parity equation remains on the other coordinates. This applies successively to $M_t$, then $L_t$ when present, then $D_t$. The finite starts contain all their own prefix submarkings, so induction beginning at $t = 6$ or $t = 8$ proves every case.

For the first pure leaf use (1). Until $M_t$ is reached, the preceding root orbit consists only of positive signed points. A signed permutation supported on such a set is the identity, since it commutes with the involution interchanging every positive and negative point. At the subsequent stages the preceding root orbit contains both signs of an initial coordinate segment. In (3), the only remaining kernel is $D_{t-1}$, supported on the first $t-1$ coordinates, and it is contained in $M_t$. In (4), the two remaining kernels are $D_{t-3}$ inside $M_t$ and $D_{t-2}$ inside $L_t$. These are exactly the supported subgroups obtained from the formulas. Thus all kernel inclusions hold. A second pure leaf is permitted by the pure-leaf corollary. The full $D_t$ is transitive on its $2t$ signed points, so (2) twice gives $S_{2t+2}$. For the last deletions, $M_t$ in (3) has root orbit of size $2t - 2$ and complement image $C_2$; $L_t$ in (4) has root orbit of size $2t - 4$ and complement image $V_4$. Applying (2) twice gives the asserted actual direct products. □

## 3 The left path and its two parity relations

Let $k \geq 1$, $m = 4k + 6$, and $L = m - 1$. On $x_1, \ldots, x_m$ put

$$\begin{aligned} a_0 &= (x_1\ x_2)(x_3\ x_4)\cdots(x_{L-2}\ x_{L-1}),\\ a_1 &= (x_2\ x_3)(x_4\ x_5)\cdots(x_{L-1}\ x_L),\\ a_2 &= (x_3\ x_4)(x_L\ x_m). \end{aligned}$$

Add two rows by defining, for $j \geq 3$,

$$a_j = (x_{m+j-3}\ x_{m+j-2})(y_{j-2}\ y_{j-1}). \tag{9}$$

These are ordinary bridges, without the final vertical modifications which will be introduced below. Write $H_j = \langle a_0, \ldots, a_j \rangle$ and $K_j = \langle a_1, \ldots, a_j \rangle$.

**Lemma 3.** *The marking of $H_2$ is C and $H_2 = \mathrm{Alt}(\{x_1, \ldots, x_m\})$. For $j \geq 3$, its extension by (9) is C, and*

$$H_j = \{(\sigma, \pi) \in S_{m+j-2} \times S_{j-1} : \varepsilon(\sigma) = \varepsilon(\pi)\}, \tag{10}$$

*on the two indicated rows. The groups $K_j$, for $j \geq 2$, are C and have the actual form*

$$\{(\sigma, \pi, d) \in S_{j+1} \times S_{j-1} \times D : \varepsilon(\sigma) + \varepsilon(\pi) = \chi(d)\}. \tag{11}$$

*Here $D = \langle a, b \rangle \cong \mathrm{Dih}(8)$ acts on $\{x_2, x_3, x_4, x_5\}$, with $a = (x_2\ x_3)(x_4\ x_5)$, $b = (x_3\ x_4)$, and $\chi(a) = \chi(b) = 1$. The first symmetric factor acts on $\{x_{m-2}, \ldots, x_{m+j-2}\}$, and the second on the bottom row. There are $2k - 1$ further two-point orbits, all interchanged by $a_1$ and fixed by the other marks. Their common action is determined by the character of $D$ taking $a$ to one and $b$ to zero.*

*Proof.* The element $(a_1a_2)^4$ is a pure three-cycle on $\{x_{L-1}, x_L, x_m\}$. The other nontrivial components of $a_1a_2$ have lengths four or two. Each of $a_0, a_1, a_2$ fixes a point of this three-element support, and their graph on the top row is connected. The acquired supported-alternating-group lemma therefore gives $H_2 = A_m$. All three marks are even. For its intersection property, $\langle a_0, a_1\rangle$ is dihedral of order $2L$ and fixes $x_m$. The group $\langle a_1, a_2\rangle$ is dihedral of order 24; its orbit at $x_m$ has three points, so that point stabilizer has order eight. Since $L$ is odd, the intersection has order dividing $\gcd(2L, 8) = 2$. It contains $a_1$, and hence is exactly $\langle a_1\rangle$. The rank-three end criterion applies.

Consider next the two right paths in $K_j$, temporarily omitting $D$ and the repeated pairs. Their marking consists of two successive pure leaves at the beginning of a diagonal adjacent-transposition path on $j-1$ paired coordinates. That diagonal path is C. Its restriction kernels on an all-positive root orbit are trivial, so the first leaf satisfies (1); the second is pure. Formula (2) shows that the resulting quotient is $S_{j+1} \times S_{j-1}$, with its stated C-marking. This includes $j = 2$, when it is just the two adjacent transpositions of $S_3$.

The two initial quotient marks are odd in total, and every subsequent paired bridge is even. This gives the relation in (11). The cube $(a_1a_2)^3$ has trivial quotient action and induces $(ab)^3$ on $D$, generating the cyclic subgroup $\ker\chi$ of order four. The repeated-pair actions are already determined by $D$. Thus the kernel is all of $\ker\chi$, and (11) is an equality. On omission of $a_1$, the $D$-action reduces to $\langle b\rangle$ and is determined by the quotient's total sign. Projection is injective there. Lemma 1 proves C for $K_j$.

To prove (10), the normal closure of the supported $A_m$ under the bridges contains the alternating group on the entire upper row. This follows from the supported-alternating-group lemma on that row: all conjugates remain supported there, the upper graph is connected, and each mark fixes a point of the initial set of $m$ points. The lower projection is $S_{j-1}$. Every even lower permutation is represented by an even-length word in its adjacent transpositions; the corresponding bridge word also acts evenly on the upper row, and that action can be cancelled using its supported alternating group. We therefore have both supported alternating factors, together with a bridge odd in both factors. This proves precisely (10).

Finally $H_{j-1}$ fixes the newly added last point on each row. Inside (11), imposing these two fixed points replaces $S_{j+1} \times S_{j-1}$ by $S_j \times S_{j-2}$, with the same $D$ relation; the result is exactly $K_{j-1}$. Consequently $H_{j-1} \cap K_j = K_{j-1}$. All $K_j$ are C and $H_2$ is C, so induction with the end criterion proves C for every $H_j$. □

# 4 Higher odd ranks

Let $r \geq 7$ be odd, $k \geq 1$, and $n = 2r + 1 + 4k$. Use the preceding marks through $a_{r-1}$ on top row $x_1, \ldots, x_{m+r-3}$ and bottom row $y_1, \ldots, y_{r-2}$, except that the mark $a_{r-3}$ is multiplied by

$$f = (x_{m+r-4}\ y_{r-3})(x_{m+r-3}\ y_{r-2}). \tag{12}$$

Call the resulting marks $\rho_0, \ldots, \rho_{r-1}$ and their group $G$. These are the literal source Theorem 4.8 diagrams. The two last vertical pairs will be denoted $Z'$ and $Z$, in that order. All marks are even involutions and distant marks commute. Write $G_I = \langle \rho_j : j \notin I\rangle$.

The right end $G_0$ has its four-point $D$-orbit, repeated pairs, and a large orbit $R$ of size $N = 2r - 2$. On $R$, its marking is (3) with $t = r - 2$, preceded by two pure leaves. Thus Lemma 2 gives the quotient $S_R$ with C. The same cube and injectivity argument as in (11) gives

$$G_0 = \{(g, d) \in S_R \times D : \varepsilon(g) = \chi(d)\}, \tag{13}$$

including the determined repeated-pair actions, and proves C. Deleting its last mark gives, from the last-deletion part of Lemma 2,

$$G_{0,r-1} = \{(g, z, d) \in S_{R\setminus Z} \times S_Z \times D : \varepsilon(g) + \varepsilon(z) = \chi(d)\}. \tag{14}$$

This is also C as a submarking of $G_0$.

We prove C for the other end $G_{r-1}$. Its deletion of the last remaining mark is $G_{r-2,r-1}$. By Lemma 3, the ordinary prefix through $a_{r-3}$ is a C-marking of the parity group on

$$U = \{x_1, \dots, x_{m+r-5}\}, \qquad V = \{y_1, \dots, y_{r-4}\}.$$

Multiplying its endpoint by the disjoint central $f$ preserves C. The adjacent preceding bond is three, so the cube separates $f$. Consequently this end is exactly

$$G_{r-2,r-1} = \{(u, v) \in S_U \times S_V : \varepsilon(u) = \varepsilon(v)\} \times \langle f \rangle. \tag{15}$$

Its intersection with (14) can now be calculated without a maximal-subgroup assertion. Put $U' = U \cap R$, of size $r-2$. An element of the intersection preserves $U', V, Z', Z$, and the actions on $Z', Z$ are the same interchange, say with exponent $e$. On the rest of $U$, the four-point $D$-action has sign equal to the $b$-character, while the odd number $2k-1$ of repeated pairs contributes the $a$-character. Their sum is $\chi(d)$. Thus (15) imposes exactly

$$\varepsilon(u|_{U'}) + \varepsilon(v) = \chi(d).$$

The relation in (14) is identical, since the two pair signs $e + e$ cancel. Therefore the intersection is

$$\{(u', v, d) \in S_{r-2} \times S_{r-4} \times D : \varepsilon(u') + \varepsilon(v) = \chi(d)\} \times \langle f \rangle. \tag{16}$$

By (11) and the same separating cube, this is exactly $G_{0,r-2,r-1}$. Both ends of $G_{r-1}$ are C, so its intersection property follows.

Finally $G_{r-1}$ preserves the last pair $Z$. The full setwise stabilizer of $Z$ in (13) is precisely (14). Hence $G_0 \cap G_{r-1} = G_{0,r-1}$, proving C for $G$. The left three marks contain the supported $A_m$ proved in Lemma 3. The full graph is connected, and every mark fixes a point of that $m$-element set. The acquired alternating support lemma gives $G = A_n$.

The adjacent orders are

$$[4k+5, 12, 6, 3^{r-7}, 6, 6, 3]. \tag{17}$$

The first is the length of the initial alternating path; the next is the least common multiple of four and three on the two nontrivial components of $a_1 a_2$. The third is a three-cycle with disjoint transpositions. The unchanged paired bridges give three, and the three final products involve only the four last paired coordinates. Their orders $6, 6, 3$ follow either directly from (3) or from its finite start. Thus these are the source's exact types as well as its rank-existence assertions.

## 5 Higher even ranks

Let now $r \geq 8$ be even, $k \geq 0$, $m = 4k+6$, and $n = 2r+3+4k$. Keep the initial marks and bridges, extend the top row to $x_{m+r-2}$ and the bottom to $y_{r-1}$, and make these two modifications:

$$\rho_{r-2} = a_{r-2}(x_{m+r-3}\ x_{m+r-2})(y_{r-2}\ y_{r-1}), \tag{18}$$

$$\rho_{r-4} = a_{r-4} \prod_{i=0}^{3} (x_{m+r-2-i}\ y_{r-1-i}). \tag{19}$$

All other marks are the corresponding $a_j$. These are the source Theorems 4.9 and 4.10 diagrams. They again give even involutions with the string commutations.

First omit $\rho_{r-1}$. There is a main component $M$ of size $n-4$ on which the marking is exactly the odd-rank construction of rank $r-1$, and a four-point component $Q$. On $Q$, only $\rho_{r-4}$ and $\rho_{r-2}$ act, as the two independent double transpositions $v, h$ of a square. The main marking is C: for $k \geq 1$ this was just proved, and for $k = 0$ it is the retained earlier maximal-rank seed of source Theorem 4.7.

Projection of $G_{r-1}$ onto $M$ is injective on the endpoint deletion omitting $\rho_{r-2}$. Indeed $h$ is then absent, and its action on $Q$ is $v^e$, where $e$ is the parity of the number of occurrences of $\rho_{r-4}$. On the main component, the last vertical pair is preserved by that endpoint deletion; exactly $\rho_{r-4}$ interchanges it. Hence $e$ is determined by the main permutation. A word acting trivially there has $e = 0$ and acts trivially on $Q$ as well. Lemma 1 proves C for $G_{r-1}$.

Its actual group is

$$G_{r-1} = \mathrm{Alt}(M) \times V_4. \tag{20}$$

For the commutator subgroup has trivial action on $Q$, and its image on $M$ is $[\mathrm{Alt}(M), \mathrm{Alt}(M)] = \mathrm{Alt}(M)$; here $|M| \geq 15$. Thus the full supported $\mathrm{Alt}(M)$ lies in the group. Cancelling the main parts of the two indicated marks then yields $v$ and $h$ separately. This proves (20).

For $k \geq 1$, the right end has the same $D$-orbit and repeated pairs as before, and a large orbit $R$ of size $N = 2r$. On $R$ the marking is (4) with $t = r - 1$, preceded by two pure leaves. Exactly as in (13), the quotient criterion and the cyclic kernel of order four give C and

$$G_0 = \{(g, d) \in S_R \times D : \varepsilon(g) = \chi(d)\}. \tag{21}$$

The last deletion, by Lemma 2, is

$$G_{0,r-1} = \{(g, d) \in S_{R \setminus Q} \times D : \varepsilon(g) = \chi(d)\} \times V_4. \tag{22}$$

Every $V_4$-element on $Q$ is even. Equations (20) and (21) therefore force their intersection to be exactly (22): preservation of $M, Q$ leaves the displayed symmetric factor, and the sign on the rest of $M$ is precisely $\chi(d)$. The end criterion proves C for the full marking when $k \geq 1$.

It remains to justify $G_0$ when $k = 0$, where the two left small orbits join and the formula with $D$ does not apply. In this case $G_0$ fixes $x_1$ and is transitive on all other $n - 1$ points. The subgroup $G_{0,1}$ has the separate pair $\{x_3, x_4\}$ and a large orbit of size $n - 4$; it fixes $x_1, x_2$. On that large orbit its marking is one pure leaf attached to (4); the first mark is multiplied by the interchange of $x_3, x_4$. It is an endpoint central twist, and its group is

$$\{g(x_3\ x_4)^{\varepsilon(g)} : g \in S_{n-4}\}. \tag{23}$$

Here only the attached leaf is odd on the large orbit and every signed-cap mark is even, which proves this precise sign relation. It contains the supported alternating group on that orbit. Every mark of $G_0$ fixes a point of the orbit, and its full graph off $x_1$ is connected. The alternating support lemma therefore gives

$$G_0 = \mathrm{Alt}(\Omega \setminus \{x_1\}). \tag{24}$$

This generation argument does not presuppose C.

For C, one end of $G_0$ is (23), already proved C. Its other end $G_{0,r-1}$ is a submarking of the C-group $G_{r-1}$. We also need its exact group. On $M$ it is the right end of the retained odd maximal-rank seed. That right end is $\mathrm{Alt}(M \setminus \{x_1\})$: the preceding argument applies verbatim with (3) in place of (4). The pure-leaf cap gives a large supported alternating group, and connectivity gives the full even group on the other points. As in (20), taking commutators and then cancelling the two square actions proves

$$G_{0,r-1} = \mathrm{Alt}(M \setminus \{x_1\}) \times V_4. \tag{25}$$

Intersect this with (23). The element fixes $x_2$, preserves $\{x_3, x_4\}$, and acts on $Q$ in $V_4$. On the remaining $n - 8$ points it is arbitrary symmetric, with its sign recorded by the interchange of $x_3, x_4$. Thus the intersection is the sign copy of $S_{n-8}$, times $V_4$. This is exactly $G_{0,1,r-1}$: use the last-deletion formula of Lemma 2 with one attached leaf and its same endpoint twist. Both ends are C and their intersection is the middle, so $G_0$ is C. Finally (24) and (20) intersect in (25), completing the full C proof.

For every $k \geq 0$, the supported $\mathrm{Alt}(M)$ in (20), the connectivity of the full graph, and a fixed point in $M$ for each mark give $G = A_n$ by the alternating support lemma. The type is

$$[5, 5, 6, 3^{r-8}, 6, 6, 6, 4] \quad (k = 0), \qquad [4k + 5, 12, 6, 3^{r-8}, 6, 6, 6, 4] \quad (k \geq 1).$$

The initial products are as above (a five-cycle when $k = 0$), the middle paired bridges give three, and the final four products are those in (4), with orders $6, 6, 6, 4$.

## 6 Scope and attribution

The written proof recovers the exact higher odd-degree families of source Theorems 4.8–4.10. The parameterization is exhaustive for their stated ranks: when $n \equiv 3 \pmod 4$, odd $r$ has $n = 2r + 1 + 4k$, and even $r$ has $n = 2r + 3 + 4k$. The odd $k = 0$ seed remains the attributed prior theorem. This note does not claim a new construction, a new open-question answer, a Lean verification, or a numerical compression factor.

The new organization consists of the two concrete signed cap calculations, the restriction-kernel transfer, and the exact parity descriptions of the end intersections. In particular, an abstract group isomorphism or a list of orbit sizes is never used as a substitute for an equality of permutation subgroups. The only new finite premises are the two complete base cap certificates. Sample computations in larger degrees are retained as diagnostics and are not premises of a uniform assertion.

This component requires distinct review before admission. The low-rank companion is already reviewed separately. A whole L038 recovery also needs the existing contraction and degree-twelve interfaces assembled in one principal scope route.

## References


[1] M. E. Fernandes and D. Leemans, *String C-group representations of alternating groups*, arXiv:1810.12450v2, 4 January 2019; Ars Math. Contemp. 17 (2019), 291–310. Theorems 4.7–4.10 and their literal diagrams.

[2] M. E. Fernandes, D. Leemans and M. Mixer, *All alternating groups $A_n$ with $n \geq 12$ have polytopes of rank $\lfloor (n-1)/2 \rfloor$*, SIAM J. Discrete Math. 26(2) (2012), 482–498. The maximal-rank construction is retained through its explicit attribution and statement in source Theorem 4.7.

[3] P. McMullen and E. Schulte, *Abstract Regular Polytopes*, Cambridge University Press, 2002, Propositions 2E16 and 2E17.